

Supernetting

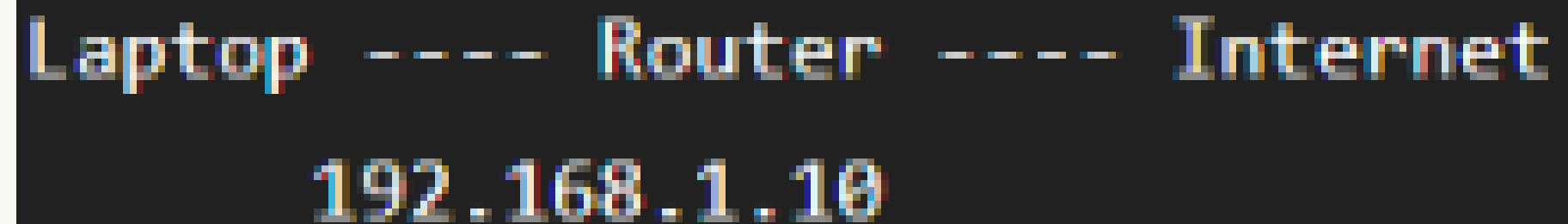
IN COMPUTER NETWORKS

A Powerpoint by

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What is an IP Address?

- Every device connected to a network needs a unique address.
- IPv4 has 32 bits.
- Example: 192.168.1.10

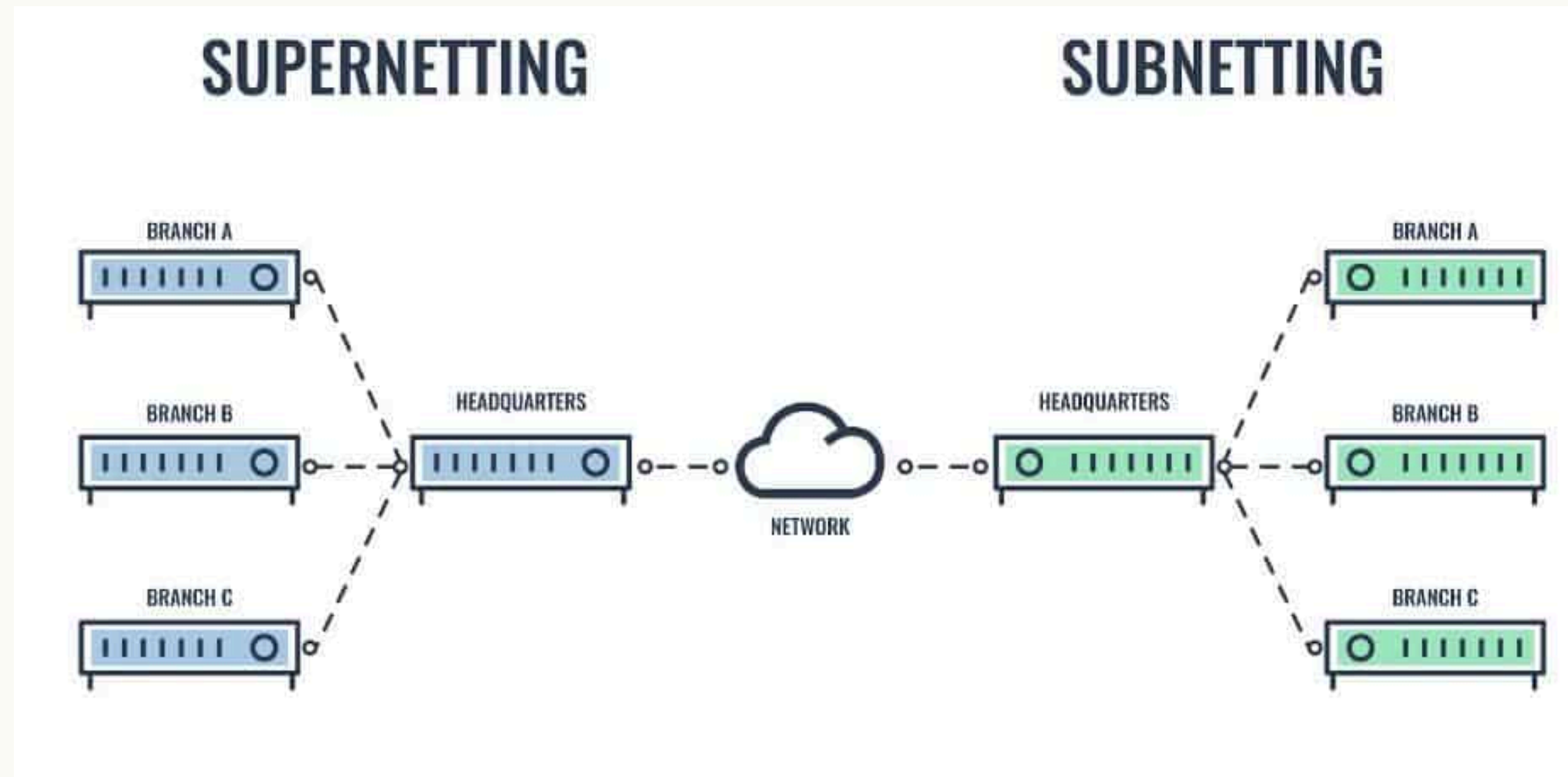


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Laptop ---- Router ---- Internet
192.168.1.10
```

Router connects multiple devices within a Local Area Network (LAN)

Why does an IP Address have 4 numbers?

- Example: 192.168.1.10
- It consists of 4 parts (octets).
 - 192 . 168 . 1 . 10
- Each part can have values from 0 to 255.
- Each part = 8 bits
- So,
 - $8 + 8 + 8 + 8 = 32$ bits
 - IPv4 Address = 32 bits



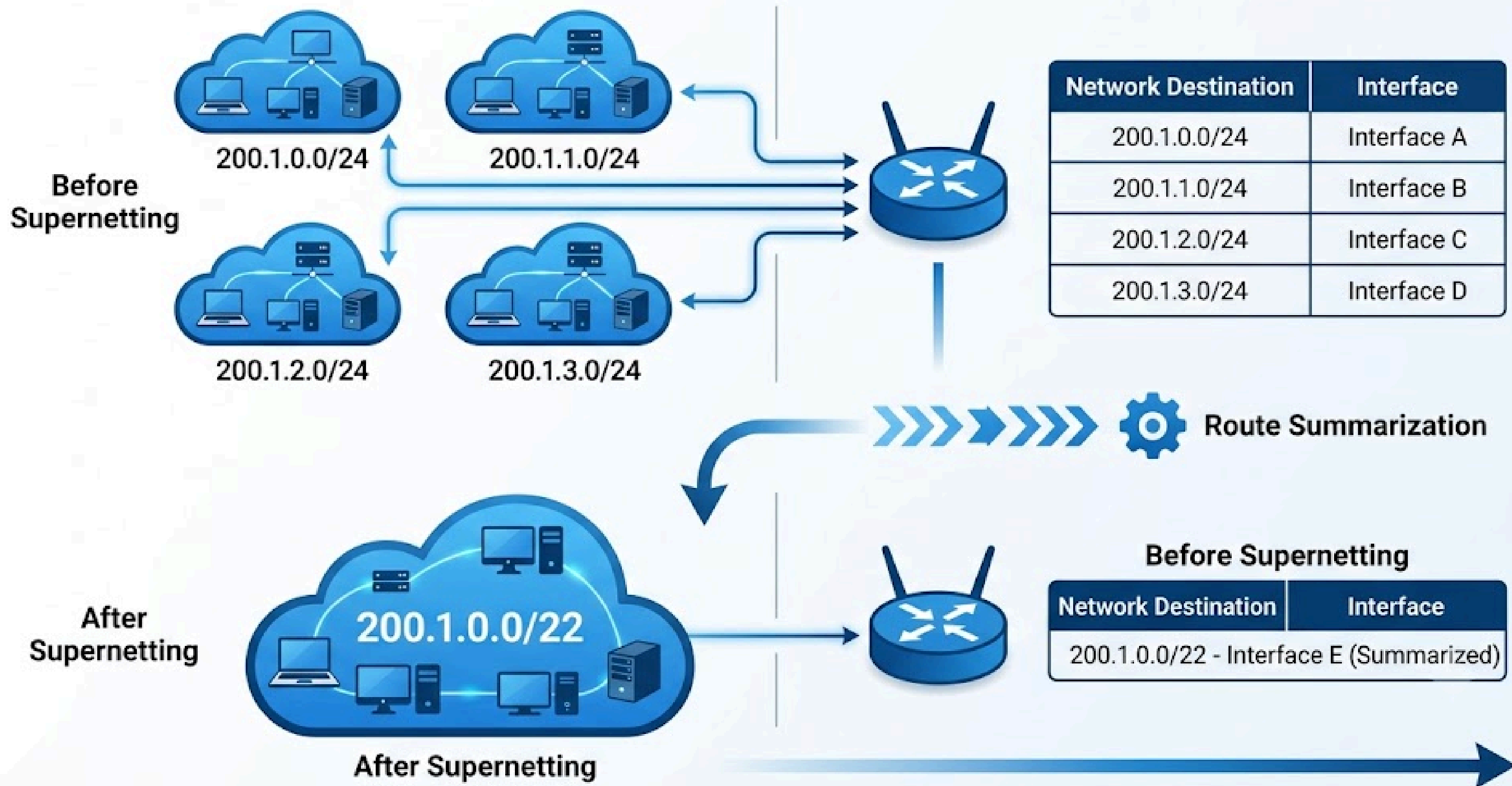
- **Subnetting** involves dividing a large IP network into smaller sub-networks.
 - This helps in organizing devices, enhancing security, and optimizing IP address allocation
- **Supernetting** combines multiple smaller networks into a larger network.
 - This technique simplifies routing and management by aggregating address spaces

What is a Routing Table?

Imagine you're a food delivery driver.



UNDERSTANDING SUPERNETTING & ROUTE SUMMARIZATION



where does /24 and /22 come from?

- The computer doesn't see it like this. It sees it in binary.

$$192 = 11000000$$

$$168 = 10101000$$

$$1 = 00000001$$

$$0 = 00000000$$

- So the full IP is:

11000000.10101000.00000001.00000000

That's 32 bits (4×8 bits).

/24 simply means:

The first 24 bits are used to identify the network.

The remaining 8 bits are for hosts.

32 bits total

<-----24 bits----->	<--8 bits-->
Network Part	Host Part

/22 (After Supernetting)

Network Bits	Host Bits
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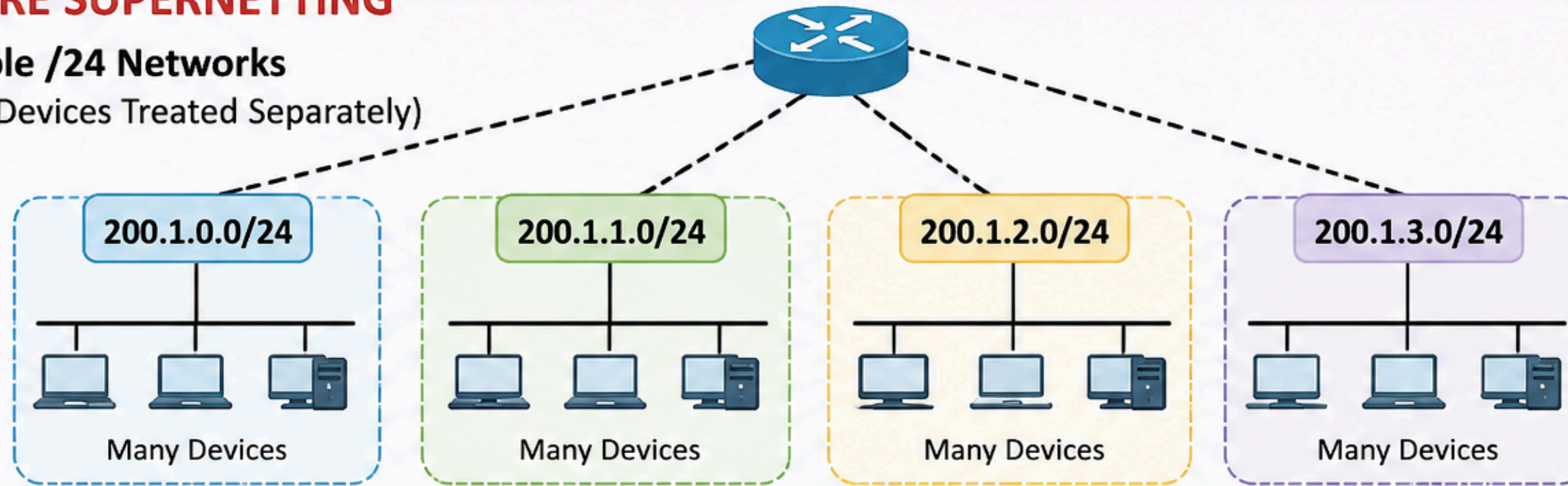
<-----22----->	<----10---->
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From /24 to /22 – Creating a Larger Space

BEFORE SUPERNETTING

Multiple /24 Networks

(Many Devices Treated Separately)



Router Sees

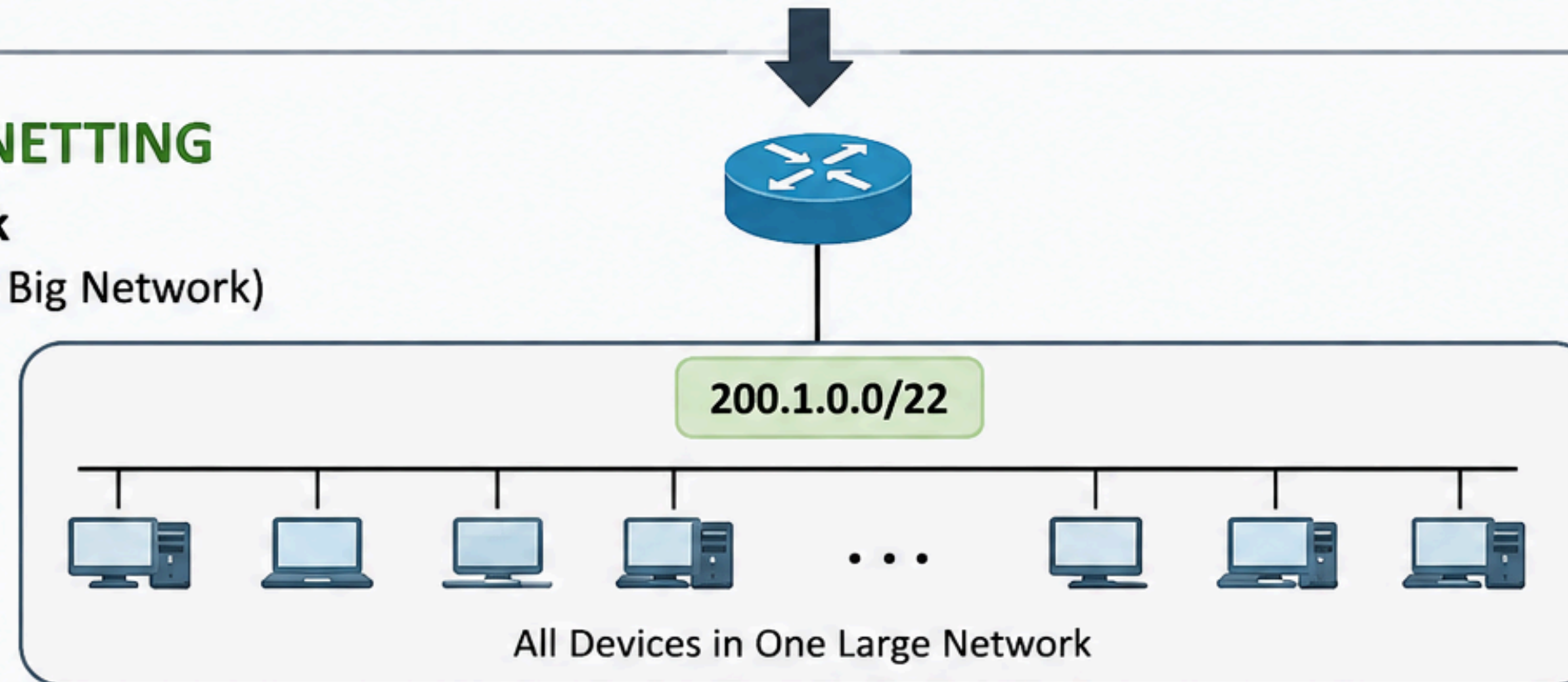
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Separate
Networks

AFTER SUPERNETTING

One /22 Network

(All Treated as One Big Network)



Router Sees

1

Single
Network



Think of it like this:

Earlier: Many houses in many streets



Now: All houses in one big city



Benefits of Supernetting:

- Saves storage space in routing tables
- Reduces the number of routing updates
- Simplifies routing decisions
- Helps manage the growth of routing tables as the Internet expands

Conditions for Supernetting

1. Networks should be Contiguous (Adjacent)

The networks must be next to each other without any gaps.

Example

 Correct

200.1.0.0/24

200.1.1.0/24

200.1.2.0/24

200.1.3.0/24

They are consecutive networks.

 Incorrect

200.1.0.0/24

200.1.2.0/24

200.1.5.0/24

200.1.7.0/24

There are missing networks between them, so they cannot be supernetted.

2. All Networks Should Be the Same Size

Every network must have the same CIDR prefix (same number of IP addresses).

Example

 Correct

200.1.0.0/24

200.1.1.0/24

200.1.2.0/24

200.1.3.0/24

 Incorrect

200.1.0.0/24

200.1.1.0/25

200.1.2.0/24

200.1.3.0/26

3. The First Network Must Start at the Correct Boundary

The first network should begin at the correct starting point for the combined block.

Example

We want to combine $4 \times /24$ networks.

The first network is:

200.1.0.0/24

This starts at 0, which is the correct boundary.

So we can combine:

200.1.0.0/24

200.1.1.0/24

200.1.2.0/24

200.1.3.0/24

↓

200.1.0.0/22

✗ But if we start from:

200.1.1.0/24

200.1.2.0/24

200.1.3.0/24

200.1.4.0/24

We cannot summarize them into a single /22, because the block doesn't start at the required boundary.

